

A TECHNICAL SEMINAR REPORT
On
ROBOTICS IN HAZARDOUS ENVIRONMENTS

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CERTIFICATE

This is to certify that this Seminar report entitled “**ROBOTICS IN HAZARDOUS ENVIRONMENTS**”, is a Bonafide Record carried out in the 8th semester submitted by **KONGA MAHADEV MITRA (21J41A05F4)** in partial fulfilment for the award of **BACHELOR OF TECHNOLOGY** in **COMPUTER SCIENCE ENGINEERING** from **Malla Reddy Engineering College**, during the academic year **2024–2025** and that this report has not been submitted elsewhere for a degree.

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ABSTRACT

Robotics plays a crucial role in hazardous environments where human presence is dangerous or impractical. These environments include disaster-stricken areas, nuclear power plants, deep-sea explorations, space missions, and chemical or biological hazard zones. Advanced robotic systems, equipped with artificial intelligence, sensors, and autonomous capabilities, enhance safety and efficiency by performing tasks such as search and rescue, environmental monitoring, material handling, and structural inspections. Innovations in robotics, including swarm robotics, soft robotics, and teleoperated systems, further improve adaptability and resilience in extreme conditions. This paper explores the latest advancements, challenges, and future directions in the deployment of robotics in hazardous environments, emphasizing their potential to save lives.

Keywords: Robotics, Hazardous environments, Autonomous robots, Remote-controlled robots, Disaster response, Safety systems, Extreme conditions, Remote sensing.

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CHAPTER – 1

INTRODUCTION

WHAT IS ROBOTICS

Robotics is a field of science which deals with design, construction and operation of robots. Industrial, medical, autonomous, humanoid robots Robotics in hazardous environments refers to the use of robotic systems to perform tasks in locations or situations that are dangerous for humans due to factors such as extreme temperatures, radiation, toxic substances, or the risk of physical harm. These environments can include places like nuclear plants, deep-sea exploration sites, space.

WHY ROBOTICS

Robotics is essential for improving efficiency, precision, and safety in various industries and applications. Robots can perform tasks that are too dangerous, repetitive, or complex for humans, making them invaluable in hazardous environments, manufacturing, healthcare, space exploration, and more. They enhance productivity, reduce human error, and operate in extreme conditions where human intervention is limited or impossible. With advancements in artificial intelligence, machine learning, and sensor technologies, robots are becoming more autonomous.

CHAPTER – 2

BACKGROUND

The development of robots for hazardous environments has evolved over decades, driven by advancements in artificial intelligence, materials science, and automation. Below is a timeline of key milestones in the evolution of robotics for hazardous environments.

1. 1950–1970s: Early Concepts and Remote-Controlled Systems

- a) **1950** – The first modern industrial robot concept, *Unimate*, was designed for automation in hazardous factory environments.
- b) **1961** – *Unimate* was deployed in General Motors' assembly line, demonstrating robotic capabilities in industrial settings.
- c) **1970s** – The U.S. military began using early bomb disposal robots, such as the *Wheelbarrow Mark I*, to handle explosives safely.

2. 1980s: Advancements in Military and Nuclear Robotics

- a) **1985** – The first robotic-assisted surgery, *PUMA 560*, laid the groundwork for robotic applications in hazardous healthcare environments.
- b) **1986** – The *Remote-Controlled Robot STR-1* was deployed at the Chernobyl disaster site for radiation monitoring and cleanup.
- c) **1989** – The U.S. military developed the *TALON* robot, which was later used in bomb disposal and hazardous material handling.

3. 1990s: Rise of Mobile and Autonomous Robots

- a) **1995** – NASA's *Sojourner* became the first robotic rover to explore Mars, paving the way for planetary exploration robots.
- b) **1997** – The *PackBot* prototype was developed by iRobot, later used for search and rescue missions.
- c) **1999** – The *Foster-Miller SWORDS* robot was introduced for military reconnaissance and combat support.

4. 2000s: Breakthroughs in Disaster and Exploration Robotics

- a) **2001** – *PackBot* was deployed for search and rescue at the World Trade Center after the 9/11 attacks.
- b) **2004** – NASA’s *Spirit* and *Opportunity* rovers landed on Mars, significantly advancing space exploration robotics.
- c) **2005** – The U.S. military deployed *BigDog*, a quadruped robot designed for carrying heavy loads in hazardous terrains.
- d) **2011** – The Fukushima nuclear disaster led to the deployment of *PackBot* and *Quince* robots for radiation monitoring and debris removal.’

5. 2010s: AI and Autonomous Robotics Revolution

- a) **2013** – Boston Dynamics introduced *Atlas*, a humanoid robot designed for disaster response and search-and-rescue missions.
- b) **2015** – The *Ocean One* humanoid underwater robot was developed for deep-sea exploration.
- c) **2018** – NASA’s *InSight* lander deployed a robotic arm to study the seismic activity of Mars.
- d) **2019** – The first AI-powered industrial inspection robot, *ANYmal*, was deployed on offshore oil rigs.

6. 2020s–Present: The Age of Smart and Collaborative Robots

- a) **2020** – Boston Dynamics’ *Spot* robot was used for pandemic-related tasks, including hospital disinfection and social distancing enforcement.
- b) **2021** – NASA’s *Perseverance* rover successfully landed on Mars, equipped with AI-driven exploration capabilities.
- c) **2022** – The *Flyability Elios 3* drone was introduced for confined-space inspections in nuclear plants and chemical factories.
- d) **2023** – The *Phoenix* underwater robot was developed for Arctic exploration, capable of operating under extreme ice-covered conditions.

CHAPTER – 3

CORE CONCEPT

Robotics in hazardous environments relies on several fundamental concepts that enable robots to function effectively in extreme conditions.

1. Autonomy and Remote Operation

- a) Robots in hazardous environments often need to function with minimal human intervention. Autonomy ranges from **remotely operated** systems, like bomb disposal robots, to **fully autonomous** robots, such as Mars rovers.
- b) NASA's *Perseverance* rover autonomously navigates the Martian surface using AI-driven decision-making.
- c) The *TALON* robot can be remotely controlled for bomb disposal in military operations.
- d) Wheeled and tracked robots, Legged robots, Aerial drones are Different locomotion techniques.
- e) **LIDAR and radar** for 3D mapping, **Thermal imaging** for detecting heat signatures, **Sonar** for underwater navigation (*ROV Jason*), **Radiation sensors** for nuclear environments

2. Artificial Intelligence (AI) and Machine Learning (ML)

- a) AI enhances robotic decision-making, allowing them to analyze data, recognize patterns, and adapt to unpredictable environments. Machine learning enables robots to improve performance over time. Inkjet-based Bioprinting: A bioprinting technique that uses droplets of bioink ejected from a print head, like inkjet printers.
- b) Boston Dynamics' *Spot* uses AI for navigation and obstacle avoidance in industrial inspections.
- c) Underwater robots like *Nereus* use AI to map deep-sea environments without human guidance.

CHAPTER – 4

APPLICATIONS

1. Disaster Response and Search & Rescue

Robots like *PackBot* and *Spot* are deployed to navigate debris and locate survivors in disaster zones, enhancing rescue operations. Their ability to reach dangerous areas ensures quicker and safer responses.

2. Nuclear and Chemical Hazard Management

STR-1 and *TALON* robots monitor radiation levels and handle hazardous materials in nuclear plants, minimizing human exposure. They perform critical decontamination and maintenance tasks in toxic environments.

3. Space Exploration and Planetary Research

Perseverance Rover and *Robonaut 2* explore Mars and assist astronauts in conducting research and maintaining equipment in space. These robots gather data from environments far beyond human reach.

4. Underwater and Deep-Sea Exploration

Ocean One and *ROV Jason* are used for underwater research and industrial inspections in deep-sea environments, where humans cannot easily go. They provide vital data and maintain underwater infrastructure.

5. Military and Defense Operations

BigDog and *SWORDS* robots support military operations by performing reconnaissance, bomb disposal, and carrying heavy loads in combat zones. They reduce the risks to human soldiers in hazardous areas.

CHAPTER – 5

METHODOLOGY

1. TECHNOLOGIES USED IN COMPUTER SCIENCE FIELD

- a) **Artificial Intelligence (AI)** – Enables robots to analyze data, make decisions, and adapt to changing environments **autonomously**.
- b) **Machine Learning (ML)** – Helps robots improve performance over time by learning from past experiences and sensor data.
- c) **LiDAR (Light Detection and Ranging)** – Provides 3D mapping and navigation for robots in complex and hazardous terrains.
- d) **RADAR (Radio Detection and Ranging)** – Assists in obstacle detection and movement tracking, especially in military and industrial robots.
- e) **Computer Vision** – Allows robots to recognize objects, assess surroundings, and navigate with precision.
- f) **Haptic Feedback** – Enables teleoperated robots to provide sensory feedback, improving control in delicate tasks.
- g) **ROS (Robot Operating System)** – An open-source framework that facilitates communication and control in robotic systems.
- h) **SLAM (Simultaneous Localization and Mapping)** – Helps robots map unknown environments while tracking their own position in real time.

2. TECHNICAL WORKFLOW IN THE BACKGROUND

- a) The operation of robots in hazardous environments follows a structured technical workflow that ensures efficiency, safety, and adaptability. This workflow includes **data acquisition, perception, decision-making, motion planning, execution, feedback**.
- b) **Data Acquisition** – Robots collect real-time data from multiple sensors such as **LiDAR, cameras, RADAR, and thermal imaging to assess the environment**.
- c) **Perception & Processing** – The sensor data is processed using AI and computer vision to recognize objects, detect hazards, and map surroundings.

- d) **Decision-Making & Planning** – AI algorithms and SLAM technology help the robot determine optimal actions, paths, or responses based on environmental conditions.
- e) **Motion Planning & Navigation** – The robot calculates safe and efficient movement paths using SLAM and autonomous navigation systems.
- f) **Task Execution** – The robot carries out the assigned task, such as debris removal, inspection, or sample collection, using robotic arms or locomotion systems.
- g) **Feedback & Adaptation** – The system continuously processes feedback from sensors to adjust movements, refine actions, and improve efficiency.

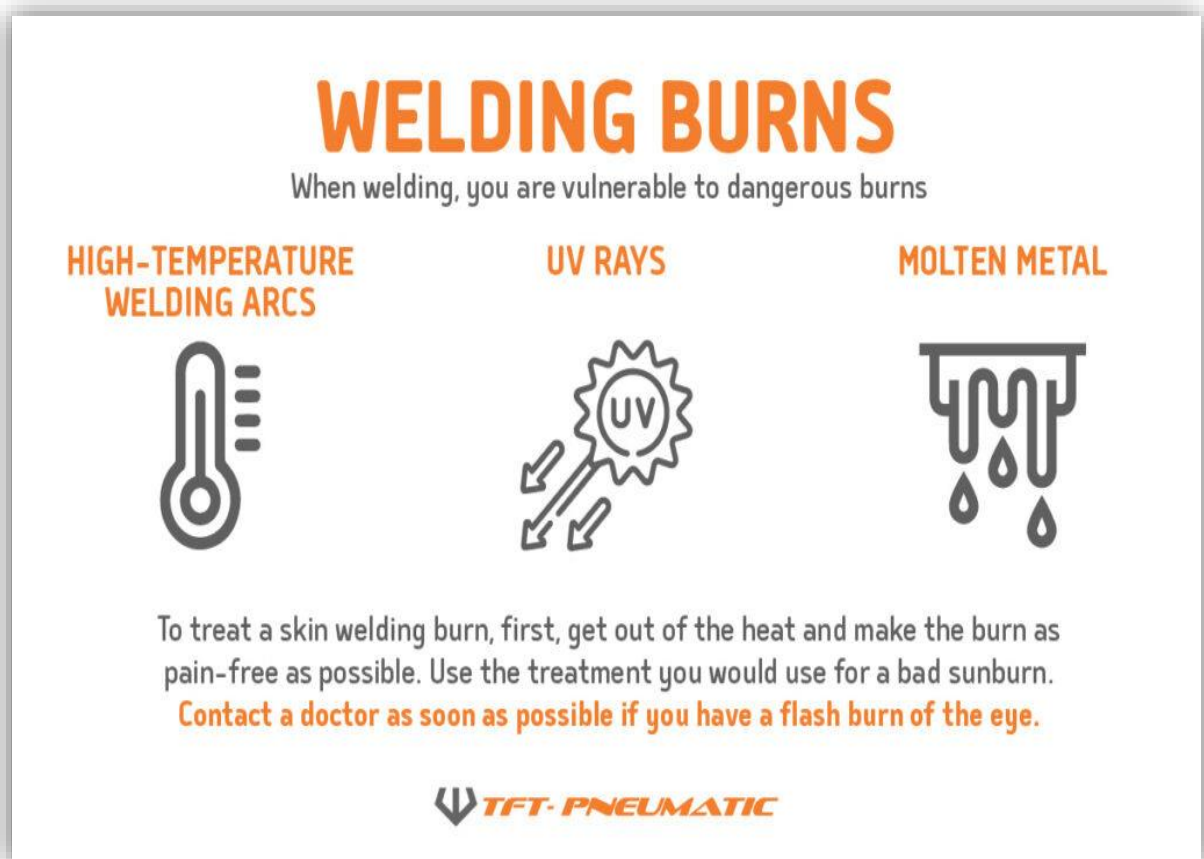
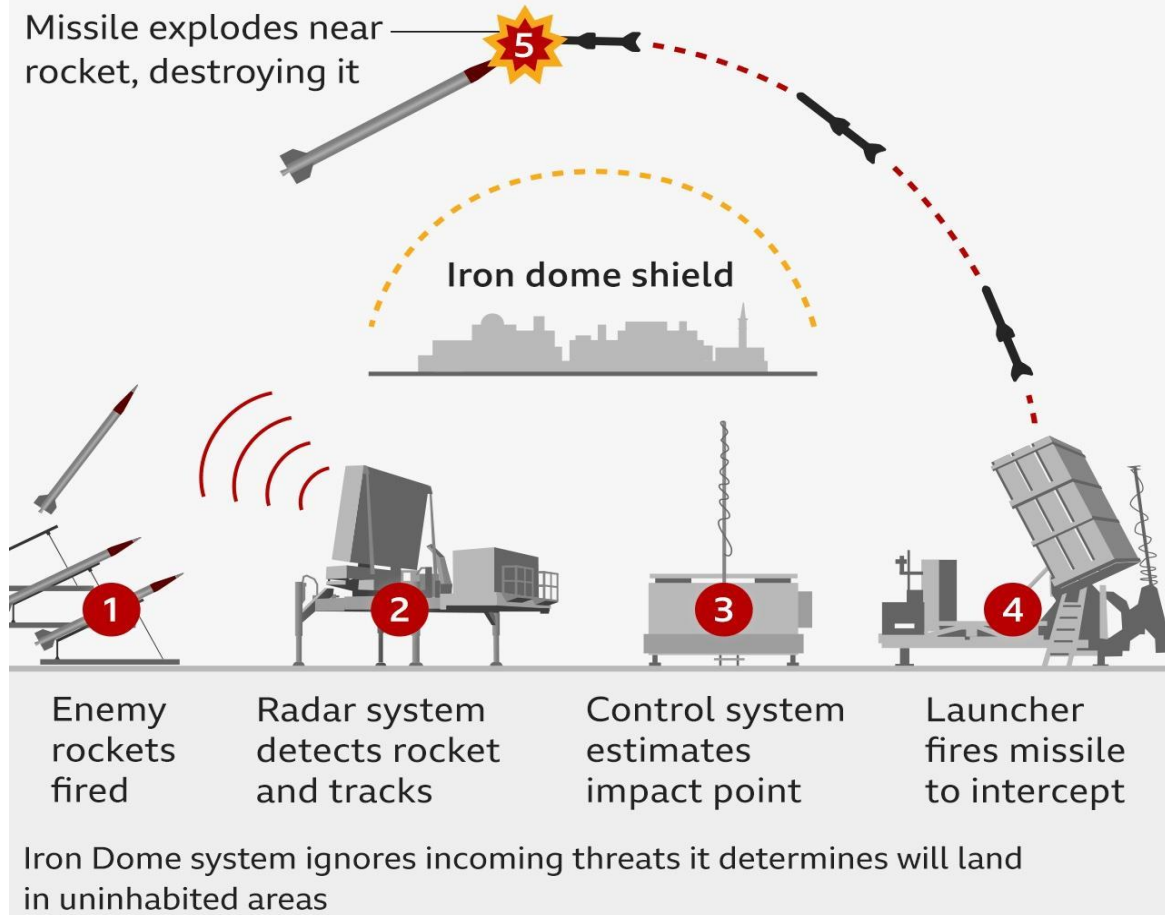


Fig 4.1 In the above figure, it shows drawbacks of humans doing welding.

How Israel's Iron Dome defence system works

Missile explodes near rocket, destroying it



Source: Rafael Advanced Defense Systems

B B C

Fig 4.2 In the below figure, it shows how Iron dome working principles.

CHAPTER – 6

ADVANTAGES AND DISADVANTAGES

ADVANTAGES OF ROBOTICS IN HAZARDOUS ENVIRONMENT

- a) Enhanced Safety :** Reduces human exposure to extreme conditions like radiation, toxic chemicals, deep-sea pressure, and war zones.
- b) Increased Efficiency :** Robots can work continuously without fatigue, improving productivity in hazardous tasks.
- c) Precision and Accuracy :** Advanced sensors and AI allow robots to perform complex operations with high accuracy.
- d) Remote Operation :** Enables human operators to control robots from a safe distance, minimizing risks.
- e) Adaptability :** AI and machine learning help robots adapt to changing environments and unpredictable situations.
- f) Cost Savings Over Time :** While initial investments are high, automation reduces long-term operational costs and risks.
- g) Data Collection & Analysis :** Robots gather real-time environmental data, improving monitoring and decision-making.

DISADVANTAGES OF ROBOTICS IN HAZARDOUS ENVIRONMENT:

- a) High Initial Costs** – Development, deployment, and maintenance of advanced robotic systems can be expensive.
- b) Complexity in Implementation** – Integrating AI, sensors, and automation requires advanced expertise and infrastructure.
- c) Limited Autonomy** – Some robots still rely on human operators for decision-making in unpredictable conditions.
- d) Risk of Malfunctions** – Harsh environments can damage robots, leading to failures and costly repairs.
- e) Ethical and Security Concerns** – Military and surveillance robots raise concerns about misuse and privacy.
- f) Job Displacement** – Automation can reduce the need for human workers in certain industries, affecting employment.

CHAPTER - 7

FUTURE SCOPE

- a) **AI-Driven Autonomy** : Future robots will rely on advanced AI and machine learning for real-time decision-making, reducing human intervention in high-risk situations.
- b) **Enhanced Mobility and Dexterity** : Robots will feature improved locomotion systems (e.g., quadrupeds, drones, and humanoid forms) for better adaptability in uneven and complex terrains.
- c) **Improved Energy Efficiency** : Advancements in battery technology and alternative power sources (such as solar and nuclear micro-reactors) will extend operational time.
- d) **Human-Robot Collaboration** : More intuitive interfaces, haptic feedback, and AI-driven communication will enhance teamwork between humans and robots.
- e) **Soft Robotics and Advanced Materials** : Robots made from flexible and self-healing materials will improve adaptability in extreme environments.
- f) **Space and Deep-Sea Exploration** : Robots will play a crucial role in planetary colonization, asteroid mining, and deep-sea resource extraction.
- g) **Self-Repairing and Self-Sustaining Systems** : AI-driven diagnostics and self-repair mechanisms will enhance durability and reduce maintenance in remote areas.

CHAPTER - 8

CONCLUSION

Robotics has revolutionized operations in hazardous environments by improving safety, efficiency, and precision in high-risk industries. From disaster response and nuclear hazard management to space exploration and military applications, robots play a crucial role in reducing human exposure to dangerous conditions. Advancements in AI, automation, and material science continue to enhance their capabilities, making them more autonomous, adaptable, and resilient. While challenges such as high costs and technical limitations remain, ongoing research and innovation promise a future where robotics will become even more integrated into critical industries. As technology progresses, robots will not only assist but also collaborate with humans, ensuring safer and more efficient solutions in hazardous environments.

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